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Data Analytics project Journal

This journal is to capture the MoM which is organised with supervisor Dr. Muhammad Adil Raja

Tags: [Data Analytics](#)

Data Analytics project progress

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This journal is to track the daily progress of the Data Analytics project:

1. **3rd Feb 2026:** Project Assignment - 30. Predicting Energy Consumption in Smart Homes using IoT Data
2. **11th Feb 2026:** Supervisor Dr Muhammad Adil Raja
3. **16th Feb 2026:** First meeting with supervisor
 1. Discussed to schedule a regular meeting i.e., every Tuesday 7:00-7:30pm.
 2. Discussed on the assigned project
 1. As I was travelling, I had a brief discussion on the plan and agreed to discuss on my research question, research progress and dataset in the next meeting.
4. **24th Feb 2026:**
 1. Discussed on the research question and agreed
 2. Discussed on the research progress:
 1. Data Cleaning, EDA
 2. What core technology will be used
 1. It was suggested to add Random forest, gradient boosting machine as feature based ML baseline and don't stick to LSTM method.
 2. For statistical model along with prophet use ARIMA, SARIMAX, VAR, ARDL

3. Discussed on the dataset and when through the UCI appliances dataset and agreed on it.
4. Discussed on the number of words interim report should have.
 1. Suggested to have at least 20-30 references.

5. **3rd March 2026:** There was no meeting due to Viva for Interim report

6. **10th March 2026:**

1. The meeting was short as I didn't had any progress on the project due to my office work.
2. As Tuesday is holiday, It was agreed the next meeting will be on Thursday 19th
3. I agreed to have progress on the Task 1 mentioned in the interim report
4. Discussed that the dataset used in this project can be used in the ML CA3. Suggestion was to check with Jack.
 1. AI - Ibrahim to check with Jack on the dataset used in both ML assignment CA3 and Data Analytics project

7. **19th March 2026:**

1. Discussed on the progress of the report.
 1. Completed Task 1 i.e. To build baseline forecasting models
2. Uploaded the code in github for review.

8. **26 March 2026:**

1. Try to add grid in the plots
2. results to be explained and make sure detail analysis is give from the plots.
3. Need to have a table comparing all the model outputs.

9. **28 April 2026**

1. The RMSE value observed are not correct for the models. For time series data, both the current value and also the previous value need to be consider.
2. Check the input data to the model, RMSE should be low
3. Validation should be done to the untested data 70 - training 15- test - 15 validation (this is un test data)

10. **9th May 2026**

1. It was discussed that RMSE value is still higher in Deep learning models
2. The aggregation is an average of 1 hour.. a concern was raised that if average is taken then it possible to loose the spikes
3. It was concluded to address it in the report and start writing the report.

11. **12th May 2026**

1. RMSE value was discussed again and it was explained that it was not an average its a sum of all values and hence the value is larger and said that its a design decision.
2. More explanation need to be given in the report.

3. Thursday meeting was said to be cancelled and the report and code will be submit

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